

SymPy Bug Card

Bug 1: solveset accepts $\operatorname{asin}(x) = \pi$

Task. Solve the principal inverse-sine equation over the complex numbers.

$$\operatorname{asin}(x) = \pi, \quad x \in \mathbb{C}.$$

SymPy's answer.

$$\operatorname{solveset}(\operatorname{asin}(x) = \pi, x, \mathbb{C}) = \{0\}.$$

Correct answer.

$$\emptyset.$$

Explanation. SymPy evaluates $\operatorname{asin}(0) = 0$, so the returned candidate is not a solution; since π lies outside the principal range of asin , the equation is unsatisfiable.

Likely root cause. In `sympy/solvers/solveset.py`, `_invert_complex` appears to apply `sin` as a global inverse of principal `asin`, without carrying the required range condition, and the finite candidate is not rejected by a residual check. This is a new instance of a known failure mode.

Artifact (GitHub). [bug-report/bug-001-solveset-accepts-asin-x-pi/](#)